

ON THE THEORY OF THE PORTEVIN-LE CHATELIER EFFECT

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As is now widely recognised a necessary condition for instability in plastic flow is that the strain rate, $\dot{\epsilon}$, should decrease with increasing stress, σ . Thus, flow will be unstable (1) if

$$\frac{d\dot{\epsilon}}{d\sigma} < 0. \quad (1)$$

Here we follow Penning (2) and suppose such instability is necessary for the appearance of the Portevin-Le Chatelier effect.

In stable flow the dependence of strain rate on stress and temperature has been successfully represented by relations of the form

$$\dot{\epsilon} = \dot{\epsilon}_0 \exp(-H/kT), \quad (2)$$

where

$$H = H(\sigma).$$

We now consider the consequences of the assumptions:

$$H = H(\sigma, C); \quad C = C(\dot{\epsilon}), \quad (3)$$

where C is the fractional occupancy by solute atoms of sites in dislocation cores.

Differentiation of eqn (2) with regard to σ then gives a relation (3,4) which can be written

$$\frac{d\dot{\epsilon}}{d\sigma} = \frac{1}{\dot{\epsilon}} + \frac{1}{kT} \frac{\partial H}{\partial C} \frac{dC}{d\dot{\epsilon}} = - \frac{1}{kT} \frac{\partial H}{\partial \sigma} = \bar{V} \quad (4)$$

where $\bar{V}$, the so-called activation volume, is necessarily a positive quantity. It follows that the inequality

$$\frac{1}{\dot{\epsilon}} + \frac{1}{kT} \frac{\partial H}{\partial C} \frac{dC}{d\dot{\epsilon}} < 0 \quad (5)$$

is a condition for unstable flow.

We now particularise. We suppose with Slezewijk (1) that dislocation glide is grossly discontinuous. Thus, a length of dislocation is supposed to be at rest for a period, t , which is large compared with that of an ensuing motion. Among other possibilities by which $H(\sigma)$ becomes $H(\sigma, C)$, we examine the case where during the periods, t , solute flows to the dislocation from the surrounding lattice as envisaged by Cottrell and Bilby (5).

For our purposes the analyses of this phenomenon for both edge (5) and screw (6) dislocations are unsatisfactory in that they allow the development with time of unlimited solute concentrations at the core. To avoid this difficulty we recognise that the rate of filling of core positions should be proportional to the ratio of the numbers of unoccupied to ultimately fillable sites $((f-C)/f)$ and simply assume that the calculated rates are reduced by this factor. Cottrell and Bilby's result then becomes, with a coefficient of solute diffusion, D ,

$$\frac{dC}{dt} = (f-C) K C_0 D^{2/3} / f b^2 t^{1/3}, D = D_0 e^{-V/kT} \quad (6)$$

$$\text{where } K = (4 A^2 / k^2 T^2)^{1/3} \quad \text{and} \quad f = C_0 e^{W/kT} / (1 + C_0 e^{W/kT}) \quad (7)$$

is the fraction (8) of occupied sites at saturation. $W = A/b$ is the interaction energy between a dislocation and a solute atom occupying a position, one-half Burger's vector distance (b) from the dislocation centre. C_0 is the initial fractional occupancy of solute sites.

We now suppose that

$$\dot{\epsilon} = \frac{N \bar{A} b}{t} \equiv \frac{\alpha}{t} \quad (8)$$

where $\bar{A}$ is the area swept by a dislocation per activated event and N is the number of active dislocation lengths. Further, we suppose that N and A are, in general, function of ϵ and T so that

$$\dot{\epsilon} = \alpha(\epsilon, T)/t \quad (9)$$

On this basis we find from eqn. (6) that, for $C_0 \ll 1$, (10)

$$C \approx f \{1 - \exp(-(3 C_0 K (D \alpha / \dot{\epsilon})^{2/3} / 2 f b^2))\} \quad (11)$$

Particularising further, we assume for simplicity that the distance between solute atoms along the dislocation core is given by the averaged quantity,

$$l = l_0 b / ((l_0 - b) C + b) \quad (12)$$

In these terms we write,

$$H = U - b l \int_0^\sigma x(\sigma) d\sigma, \quad l \neq l_0 \quad (13)$$

where $x(\sigma)$ is the distance through which the force $\sigma b l$ is moved in the activation process.

Substituting from eqns. (9), (11), (12) and (13) in eqn. (5) we obtain as a condition for instability,

$$1 \approx \frac{2 \beta f b^2}{3 k T} \frac{\exp(-\beta) l_0 (l_0 - b) \int_0^\sigma x(\sigma) d\sigma}{\{(l_0 - b)(f(1 - \exp(-\beta) + b))\}^2} = \gamma \quad (14)$$

with

$$\beta = \frac{3}{2} \frac{K C_0}{f b^2} \left(\frac{D \alpha}{\dot{\epsilon}} \right) \quad (15)$$

In view of the exponential dependence of D on $1/T$ it is clear from inspection that γ is essentially constant for such variations in $\dot{\epsilon}$ and T as make,

$$\beta \sim \text{constant.} = \bar{\beta} \quad (16)$$

Furthermore, it is readily shown that γ has one extremum, a maximum,

$$\gamma_m \approx \frac{l_0 b \int_0^\sigma x(\sigma) d\sigma}{6 k T} \quad (17)$$

when*

$$\beta (l_0 - b) f \approx b; \quad b \ll l_0 - b \quad (18)$$

These conclusions are subject to the physically imposed proviso that the rest time, $t = \alpha / \dot{\epsilon}$, is large enough for a solute atom to move, on average, at least one atom spacing.

*This small value for β is a consequence of the oversimplified representation of l employed.

Clearly, if γ reaches a value, $\gamma_m > 1$, it achieves the value unity for two values of β . We then expect a zone of plastic instability in $\dot{\epsilon} - 1/T$ space. To delineate this zone we suppose that α is only a weak function of temperature and then distinguish two extreme cases: low temperature, $f \approx 1$; high temperature, $f \approx C_0 \exp(W/kT)$. For these specifications the straight lines in $\ln \dot{\epsilon} - 1/T$ which bound the zone are from eqns. (8) and (15):

$$\ln \dot{\epsilon} = \frac{3}{2} \ln \left[\frac{3KC_0 \alpha^{2/3}}{2b^2 \beta_1} \right] + \ln D_0 - \frac{V}{kT} \quad (19)$$

$$\ln \dot{\epsilon} = \frac{3}{2} \ln \left[\frac{3K\alpha^{2/3}}{2b^2 \beta_2} \right] + \ln D_0 - \frac{V}{kT} - \frac{3W}{2kT} \quad (20)$$

In view of the approximations of the foregoing analysis and of the uncertainty regarding the magnitude and temperature dependence of α , only a qualitative comparison of these results with experiment is appropriate. In this four features are notable:

- (a) The zone of instability in $\ln \dot{\epsilon} - 1/T$ space is in fact invariably bounded by straight lines of slope differing by amounts comparable with those to be expected for $3W/2$;
- (b) and (c) as noted by Nakada and Keh (8) the "onset" of instability is dependent (c.f. eqn. (9)) on solute concentration and its disappearance is independent (c.f. eqn. (20)) of this quantity;
- (d) the quantity, V , is usually equal to the activation energy for solute diffusion when the solute is interstitial and to about half that value when it is substitutional.

The dependence of "onset" of instability on the three halves power of the concentration (Nakada and Keh give : $C_0^{1.46}$) is significant because this behaviour can be traced to the kinetics of the drift of solute to a dislocation. As might then be expected this concentration dependence is not predicted from (incomplete) analyses analogous to that given here but severally based on Schoek-Seeger hardening (9) and on the supply of solute by dislocation pipe diffusion (10).

Theory and experiment (8) deviate, however, in that the apparent activation energy for interstitial nitrogen in nickel is about half of that for diffusion. Towards an explanation of this result Nakada and Keh (8) noted the suggestion (11) that anomalously low values of the activation energy for diffusion could arise from an association of nitrogen with vacancies introduced during plastic deformation. They rejected this possibility because the observed mechanical behaviour was independent of specimen quenching history. But this observation may not be conclusive for it is readily shown that most (thermally induced) vacancies will drift to dislocations during quenching at normal rates when the activation energy for diffusion is as low as 140kJ/mol (half energy for self-diffusion) and when dislocation density is $\sim 10^9$ lines/cm². Such vacancies would then be localised and irrelevant here, irrespective of whether or not they are actually absorbed by the (partial) dislocations.

The present analysis admits of another possible explanation for such anomalies: α (eqn. (9)) can be a strong rather than a weak function of stress and temperature. An investigation of this possibility taking coquizzance of the stress dependence of $\bar{A}$ (13) is incomplete.

Conclusions

Using the Cottrell-Bilby model for the drift of solute to a dislocation one can obtain equations for both the "onset" and "disappearance" of unstable plastic flow which accord broadly with experiment.

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